

High Availability — Guided Notes

Unit 3.2.1 | CompTIA Network+ N10-009 Obj. 3.2 | N10-008 3.2

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

High availability is measured as a percentage of uptime. 'Five nines' (99.999%) allows approximately _____ minutes of downtime per year. Why do small improvements in availability percentage represent disproportionately large reductions in downtime?

Question 2

HSRP (Hot Standby Router Protocol) and VRRP (Virtual Router Redundancy Protocol) both provide _____ router redundancy by presenting a shared _____ IP and MAC address to clients. What happens to client traffic if the active router fails?

Question 3

In a redundant network design, a _____ point of failure is a single device or link whose failure causes a complete outage. Name two common single points of failure in a school network and how redundancy eliminates each.

Question 4

EtherChannel (LACP) bundles multiple physical links into one logical link, increasing _____ and providing _____. If one physical link in a four-link bundle fails, what percentage of bandwidth is retained?

Question 5

A UPS (Uninterruptible Power Supply) provides battery power during an outage long enough for a _____ shutdown or _____ activation. Without a UPS, what specific type of data corruption risk exists when a switch or server loses power mid-operation?

Question 6

Recovery Time Objective (RTO) defines the maximum acceptable _____ after a failure. Recovery Point Objective (RPO) defines the maximum acceptable _____ of data. A school that backs up data nightly has an RPO of approximately _____ hours.

Question 7

Spanning Tree Protocol (STP) prevents _____ in redundant switched networks by blocking one or more redundant ports. The tradeoff with STP redundancy is that blocked ports require _____ to transition to forwarding if the active path fails.

Question 8 — Real World Application

REAL WORLD: A school's entire network goes down when a single uplink cable is accidentally disconnected by a custodian. The outage lasts 45 minutes while staff locate a replacement cable. Identify the design flaw, name the redundancy solution, and describe what the failure experience would look like with that solution in place.

Download the answer key from the class server:

```
wget http://[SERVER]/resources/network-engineer/3-2-1-high-availability/3-2-1-high-availability-answer-key.pdf
```